

A Remarks on the Time-Token View

The transformation from the variable-token representation $\mathbf{H}_{\text{var}}^{\text{ts}}$ to the time-indexed representation $\mathbf{H}_{\text{sem}}^{\text{ts}}$ should be interpreted as a consistent readout mechanism rather than a strict inverse of inverted tokenization.

Specifically, this transformation does not attempt to reconstruct explicit temporal tokens from variable tokens in a reversible manner. Instead, it exposes an indexable axis of length L that enables step-wise alignment objectives between time-series representations and timestamped textual signals.

Such a view transformation should therefore be regarded as a lightweight structural adaptation that facilitates alignment objectives, rather than an architectural component responsible for modeling temporal dependencies.

Temporal structure is primarily modeled by the upstream Transformer encoder, while the readout mapping only provides a convenient representation for cross-modal interaction.

B Diagnostic Analyses

This appendix provides additional diagnostic analyses that support the regime-based interpretation discussed in the main paper. We analyze the interaction between forecasting and alignment objectives through gradient consistency and evaluate robustness under noisy numerical inputs.

B.1 Gradient Consistency

To examine whether the alignment objectives cooperate with the forecasting optimization, we measure the *gradient consistency* (GC) between the prediction loss and an alignment loss using the cosine similarity between their gradients:

$$\text{GC} = \cos(\nabla_{\theta}\mathcal{L}_{\text{pred}}, \nabla_{\theta}\mathcal{L}_{\text{align}}) = \frac{\langle \nabla_{\theta}\mathcal{L}_{\text{pred}}, \nabla_{\theta}\mathcal{L}_{\text{align}} \rangle}{\|\nabla_{\theta}\mathcal{L}_{\text{pred}}\| \|\nabla_{\theta}\mathcal{L}_{\text{align}}\|}. \quad (25)$$

Here θ denotes the trainable parameters of the model. $\mathcal{L}_{\text{pred}}$ is the forecasting objective (MSE), and $\mathcal{L}_{\text{align}}$ represents an alignment loss (SA, MA, or DA). Positive GC values indicate cooperative optimization directions, while negative values indicate gradient conflict.

We compute GC per parameter tensor and visualize the resulting distributions as histograms.

SA-dominant case. On TCAP-FM, the GC distribution of SA-only remains relatively compact and mildly positive, indicating stable cooperation between semantic alignment and forecasting optimization. In contrast, MA-only exhibits a broader distribution with higher variance, suggesting less consistent optimization behavior.

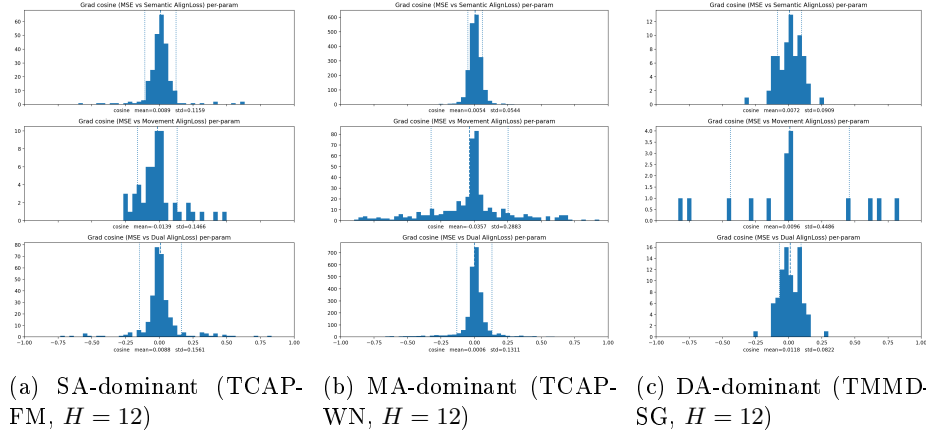


Fig. 2: Per-parameter gradient cosine similarity (GC) between forecasting loss (MSE) and alignment losses. Each panel reports three variants (top: SA-only; middle: MA-only; bottom: DA).

MA-dominant case. On TCAP-WN, MA-only shows stronger but higher-variance interactions with the forecasting objective, indicating that movement alignment can induce larger updates while occasionally conflicting with forecasting gradients. SA-only remains comparatively weak in this setting.

DA-dominant case. On TMMD-SG (Time-MMD SocialGood), dual alignment produces a more compact and positively shifted GC distribution compared with MA-only, suggesting that combining SA and MA stabilizes alignment updates and improves cooperation with forecasting optimization.

Figure 2 visualizes representative GC distributions for these regimes.

B.2 Noise Injection Robustness

To evaluate robustness under corrupted numerical observations, we inject Gaussian noise into the numerical inputs:

$$\mathbf{X}^{(\eta)} = \mathbf{X} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}\left(\mathbf{0}, (\eta \cdot \text{std}(\mathbf{X}))^2\right), \quad (26)$$

where η controls the noise level and $\text{std}(\cdot)$ is computed along the time dimension.

The forecasting error under noise is evaluated as

$$\text{MSE}(\eta) = \mathbb{E} \left[\left\| \hat{\mathbf{Y}}(\mathbf{X}^{(\eta)}, \mathbf{S}) - \mathbf{Y} \right\|_2^2 \right]. \quad (27)$$

As η increases, forecasting performance gradually degrades for all methods, but the degradation rates vary across alignment designs.

We observe three representative robustness regimes:

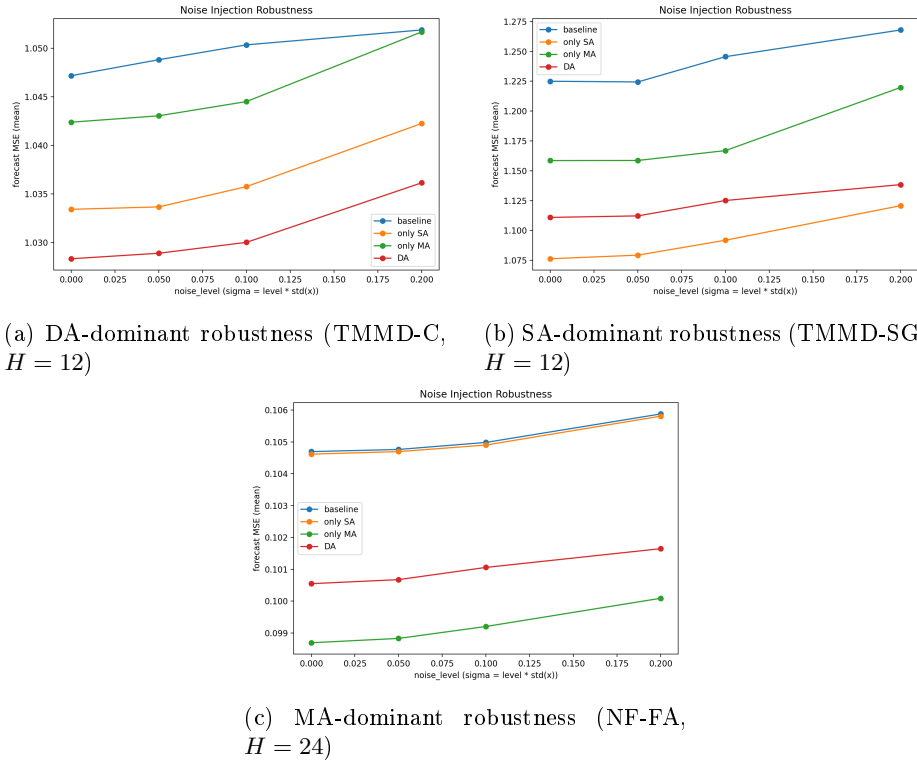


Fig. 3: Noise injection robustness under Gaussian perturbations with $\sigma = \eta \cdot \text{std}(\mathbf{X})$.

- **DA-dominant robustness.** On TMMD-C (Time-MMD Climate), dual alignment consistently achieves the lowest MSE across noise levels, suggesting that combining semantic and movement constraints helps preserve predictive stability.
- **SA-dominant robustness.** On TMMD-SG, SA-only remains the most robust across noise levels, indicating that semantic regularization alone can effectively stabilize text-conditioned representations.
- **MA-dominant robustness.** On NF-FA (NewsForecast-Finance-Australia), movement alignment is the most resilient to noise perturbations. Interestingly, MA-dominant robustness mainly appears in multivariate forecasting settings, suggesting that preserving dynamic structural patterns becomes particularly beneficial when multiple variables interact.

Overall, these results indicate that alignment mechanisms improve robustness through complementary mechanisms: semantic alignment stabilizes cross-modal semantics, movement alignment preserves dynamic structure, and their combination can further enhance robustness in certain regimes.

C Additional Transfer Results

To further examine whether the proposed dual alignment (DA) mechanism can generalize beyond the DA-Transformer architecture, we transplant the DA objectives and modules onto several representative forecasting backbones. The resulting models are denoted with the suffix `_DA`.

Table 3 reports the detailed dataset-level results comparing each backbone with its DA-enhanced counterpart. In the table, bold indicates the overall best result (SOTA) across all methods, while underline indicates the second-best result. For each DA variant, the symbols $\uparrow$ and $\downarrow$ indicate whether the corresponding metric improves or degrades relative to its original backbone.

The results show that DA-enhanced models frequently outperform their original counterparts across multiple datasets and architectures. Although improvements are not uniform across all settings, DA variants achieve new SOTA results on several benchmarks, suggesting that the proposed alignment mechanism is not tied to a specific architecture and can serve as a generally applicable design component for multimodal time-series forecasting models.

C.1 Transfer Metrics Details

This appendix reports the detailed statistics used to evaluate the transfer effectiveness of dual alignment across different backbone architectures.

The following metrics are used:

- **All-metric Win Rate (AWR)**: percentage of datasets where both MSE and MAE improve after introducing DA.
- **Partial-metric Win Rate (PWR)**: percentage of datasets where at least one metric improves.
- **Mean Relative Improvement (MRI)**: average relative improvement over improved metrics.
- **Mean Relative Degradation (MRD)**: average relative degradation over degraded metrics.

The aggregated statistics across forecasting horizons $H = 12$ and $H = 24$ are summarized in Table 4.

Table 3: Performance of DA-enhanced baselines (MSE/MAE), with SOTA reference from Table 1.

Data-set	Metr.	MM-T. (iTr.) DA	MM-T. (iTr.)	MM-T. (Pat.) DA	MM-T. (Pat.)	NLin. _DA	NLin.	MaTr. _DA	MaTr.	Uni. _DA	Uni.	T-LLM DA	T-LLM	SOTA in Tab.1
TMMD-H	MSE	1.3095↓	1.2430	1.1993↓	1.1842	1.2483↑	1.2869	1.1750↑	1.2000	1.8822↑	1.9075	2.2902↓	2.2487	1.0829
	MAE	0.6694↓	0.6653	0.6964↑	0.7005	0.7582↑	0.8114	0.7145↑	0.7234	0.8900↑	0.9025	1.1089↓	1.0309	0.6747
TMMD-A	MSE	0.4445↓	0.4299	0.4189↑	0.4513	0.4006↑	0.4854	1.0370↑	1.1749	1.2982↓	1.0529	0.5521↓	0.5517	0.4329
	MAE	0.4576↑	0.4718	0.4527↑	0.4900	0.4323↑	0.4686	0.7338↑	0.7643	0.8588↓	0.7848	0.5761↑	0.5896	0.4554
TMMD-C	MSE	0.6652↑	1.1604	1.1264↓	1.0239	1.1180↓	1.0645	1.1239↑	1.2241	1.1145↑	1.2081	1.2198↓	1.1798	1.0053
	MAE	0.9016↓	0.8984	0.8795↓	0.8329	0.8473↓	0.8239	0.8492↑	0.8815	0.8340↑	0.8782	0.8778↓	0.8663	0.8096
TMMD-Ec	MSE	0.0140↑	0.0164	0.0173↑	0.0181	0.0401↑	0.0448	0.7890↑	1.1295	3.4598↑	5.9978	0.0473↓	0.0406	0.0145
	MAE	0.0933↑	0.1035	0.1056↑	0.1086	0.1658↑	0.1685	0.8422↑	1.0094	1.6999↑	2.4338	0.1774↓	0.1655	0.0952
TMMD-En	MSE	0.1744↑	0.1825	0.1643↑	0.1644	0.1494↑	0.1654	0.2071↑	0.2985	0.7198↑	1.0659	0.3495↓	0.3272	0.1588
	MAE	0.2959↑	0.3049	0.2880↑	0.2943	0.2788↑	0.2979	0.3421↑	0.4101	0.6427↑	0.8020	0.4526↓	0.4401	0.2898
TMMD-Env	MSE	0.4101↓	0.4042	0.4308↓	0.4124	0.4227↓	0.4117	0.3978↑	0.4310	0.5072↑	0.5555	0.4605↑	0.4635	0.3988
	MAE	0.4726↓	0.4646	0.4672↓	0.4621	0.4680↑	0.4685	0.4638↑	0.4876	0.5493↑	0.5915	0.4880↓	0.4823	0.4612
TMMD-S	MSE	100.0897↑	101.6838	99.6815↑	100.6717	98.0014↑	94.9721	113.1712↓	112.9370	114.6106↑	114.6854	103.6998↓	95.1019	99.4314
	MAE	5.2849↓	5.2649	5.2214↓	5.1875	4.4651↑	4.5411	6.3443↓	6.3201	6.4487↑	6.4580	5.4526↓	4.8132	5.2426
TMMD-SG	MSE	1.4173↓	1.2973	1.3202↑	1.4221	1.3917↓	1.3880	1.0070↓	0.8662	1.5128↑	1.7966	1.9071↑	1.9108	1.2297
	MAE	0.6702↓	0.5762	0.6503↑	0.7207	0.7101↑	0.8353	0.5838↑	0.6282	0.8773↑	1.1014	1.0982↑	1.1052	0.6408
TMMD-T	MSE	0.2556↓	0.2291	0.2091↓	0.2036	0.2923↓	0.2651	0.2849↑	0.2857	0.6353↑	1.4763	0.3853↓	0.3720	0.1988
	MAE	0.2764↓	0.2524	0.2390↓	0.2321	0.3799↓	0.3754	0.4308↑	0.4331	0.6542↑	1.1122	0.4413↓	0.4352	0.2321
NF-FA	MSE	0.0729↑	0.1118	0.0670↑	0.1134	0.0701↑	0.1326	0.8228↑	2.1544	2.0969↑	3.6883	0.1496↑	0.1498	0.0727
	MAE	0.1140↑	0.2198	0.0998↑	0.2226	0.1260↑	0.2537	0.5824↑	1.1270	0.9859↑	1.5509	0.2129↑	0.2151	0.1086
NF-FC	MSE	0.6561↓	0.4766	0.6092↑	1.1376	0.5150↑	1.0733	0.4389↓	0.4196	0.4461↑	0.6785	0.5843↑	0.5868	0.5681
	MAE	0.5532↑	0.5562	0.5433↑	0.5521	0.5316↑	0.5355	0.5574↑	0.5702	0.6060↓	0.6054	0.5284↑	0.5275	0.5457
NF-T	MSE	1.2444↓	1.2313	1.2294↓	1.2250	1.2253↑	1.2301	0.5751↑	0.6257	1.2749↑	1.3301	1.3415↑	1.3427	1.2264
	MAE	0.7810↓	0.7787	0.7763↓	0.7738	0.7837↑	0.7914	0.4112↑	0.4410	0.7979↑	0.8196	0.8311↑	0.8299	0.7675
TCAP-WH	MSE	0.8369↓	0.8112	0.8060↓	0.7964	0.7960↓	0.7793	0.7980↑	0.8941	0.9339↑	0.9696	0.8197↑	0.8159	0.8150
	MAE	0.6732↓	0.6630	0.6609↓	0.6571	0.6559↓	0.6488	0.6457↑	0.7083	0.7424↑	0.7696	0.6688↓	0.6672	0.6662
TCAP-WN	MSE	0.7556↓	0.7508	0.7376↓	0.7363	0.7378↓	0.7289	0.7946↑	0.8871	0.8966↑	0.9759	0.7649↓	0.7615	0.8134
	MAE	0.6679↓	0.6659	0.6618↓	0.6612	0.6605↑	0.6605	0.6934↑	0.7442	0.7491↑	0.7982	0.6805↓	0.6786	0.6834
TCAP-WS	MSE	0.7918↑	0.8006	0.8060↓	0.8014	0.7970↑	0.8084	0.9592↑	1.1199	1.1272↑	1.3613	0.8623↑	0.8601	0.7884
	MAE	0.6546↓	0.6542	0.6590↓	0.6522	0.6567↓	0.6562	0.6817↑	0.7588	0.7771↑	0.8638	0.6804↑	0.6792	0.6548
TCAP-FM	MSE	0.0900↑	0.1579	0.0937↑	0.1453	0.0922↑	0.1567	1.8272↑	2.6243	2.6846↑	4.4568	0.2154↓	0.2087	0.1008
	MAE	0.2328↑	0.2938	0.2237↑	0.2826	0.2234↑	0.2892	1.0387↑	1.1054	1.0304↑	1.4979	0.3433↓	0.3349	0.2342
TCAP-HM	MSE	0.4543↓	0.4226	0.4242↓	0.4240	0.4580↑	0.5744	0.7050↑	0.7633	0.6116↓	0.5743	0.6635↑	0.6675	0.3887
	MAE	0.4857↓	0.4622	0.4548↑	0.4680	0.5172↑	0.5797	0.7048↑	0.7639	0.5330↑	0.5792	0.5840↑	0.6040	0.4573
TCAP-HP	MSE	3.6938↑	3.7627	3.4469↓	3.4236	3.8359↑	4.0059	5.7147↑	6.5429	6.4403↓	6.3504	4.4794↑	4.3463	3.4236
	MAE	0.8539↑	0.8935	0.8444↓	0.8391	0.8900↑	0.9988	1.3323↑	1.3960	1.4555↓	1.4169	1.0177↓	1.0041	0.9107
TTC-C	MSE	0.6568↓	0.6483	0.6372↓	0.6291	0.6420↓	0.6359	0.5957↓	0.5901	0.7403↑	0.7735	0.6732↑	0.6656	0.6349
	MAE	0.5791↑	0.5801	0.5722↓	0.5669	0.5763↑	0.5844	0.5713↓	0.5642	0.6496↑	0.6758	0.5944↓	0.5914	0.5669
TTC-M 7679	MSE	1.6796↑	2.0028	1.3059↑	1.6166	1.1456↑	2.0776	3.7963↑	4.7009	4.8369↓	4.8152	2.9614↑	2.9943	1.5711
	MAE	0.8913↑	0.9633	0.7815↑	0.8590	0.7531↑	0.9612	1.3349↑	1.5874	1.7394↑	1.7428	1.1198↑	1.1264	0.8631
TTC-M 8355	MSE	3.4697↓	3.4278	3.2362↓	3.1111	2.8149↑	3.1214	3.7031↓	3.2655	3.0296↑	3.2098	3.4765↓	3.3953	3.0221
	MAE	1.4331↓	1.4224	1.3593↓	1.3358	1.2760↑	1.3374	1.4835↓	1.3630	1.2891↑	1.3245	1.4158↓	1.3967	1.2955
TTC-M 15432	MSE	1.5204↑	1.6077	1.4684↑	1.5335	1.2960↑	1.6863	2.7163↑	2.9719	3.6384↑	3.8612	1.6962↓	1.7278	1.4930
	MAE	0.9046↑	0.9446	0.8429↑	0.9203	0.8279↑	1.0290	1.2725↑	1.3949	1.5331↑	1.5907	1.0140↑	1.0202	0.8503
TTC-M 20415	MSE	4.1661↑	4.1907	4.0806↓	3.9465	3.7047↑	4.0105	4.3156↑	4.6530	4.2608↑	4.4719	4.0463↑	4.0384	3.8226
	MAE	1.1908↑	1.1986	1.1723↓	1.1328	1.1079↑	1.1578	1.2803↑	1.3674	1.3527↑	1.4090	1.1939↑	1.1906	1.1376
TTC-M 24501	MSE	1.3523↓	1.2793	1.3702↑	1.3914	0.9298↑	1.0250	1.5697↑	1.6490	2.2317↑	2.5287	1.6263↑	1.6894	1.0277
	MAE	0.7882↓	0.7746	0.8213↑	0.8426	0.6208↑	0.7127	0.9753↑	1.0592	1.2346↑	1.3292	0.9184↑	0.9509	0.7126

Table 4: Transfer effectiveness metrics across forecasting horizons (H=12, 24)

Horizon	Metr.	MM-T. (iTrans.)	MM-T. (Patch.)	NLin.	MaTrans.	Uni.	Time-LLM
H=12	AWR	41.67%	45.83%	70.83%	79.17%	83.33%	33.33%
	PWR	58.33%	58.33%	87.50%	79.17%	91.67%	62.50%
	MRI	14.08%	14.00%	15.34%	16.85%	27.01%	1.55%
	MRD	-3.88%	-3.10%	-2.99%	-11.88%	-0.11%	-6.12%
H=24	AWR	42.11%	42.11%	36.84%	42.11%	26.32%	10.53%
	PWR	47.37%	47.37%	68.42%	57.89%	57.89%	31.58%
	MRI	11.21%	9.98%	11.31%	12.75%	4.08%	2.37%
	MRD	-8.01%	-4.42%	-6.17%	-12.42%	-24.25%	-6.04%